

SOFTWARE CHANGE IMPLEMENTATION REPORT

COURSE MANAGEMENT SYSTEM

Change Request: CR004
Global Session Expiry and Friendly Re-Login

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[Software Change Management / ICT]
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[Institution / Department]

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SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-004: Student Input Validation and Strong Password Enforcement

1. 1. Change Request Summary

Field	Details
Change Request ID	CR-004
Change Request Title	Student Input Validation and Strong Password Enforcement
Priority	High
Change Type	Frontend and Backend Validation / Security Enhancement
Affected Module	Student Management
Implementation Status	Completed
Date	8 October 2026

1.1 Business Requirement

The Student Management System must ensure that student information entered by an administrator is valid, consistent, and secure. The system must prevent invalid registration numbers, names, email addresses, and phone numbers from being stored. New student accounts must also use strong passwords to reduce the risk of unauthorized access.

1.2 Existing Behaviour

Before this change, the student form mainly checked whether required fields were filled. The system did not provide complete field-level validation for name length and characters, email format, phone number format, or password strength. Backend validation was also limited to required-field checks and duplicate registration number/email checks.

1.3 Requested Change

- Validate the registration number before submission.
- Require first name and last name to contain at least 3 characters.
- Allow valid name characters and reject inappropriate characters.
- Validate the email address format.
- Require the phone number to contain exactly 10 numeric digits.
- Require strong passwords for new student accounts: minimum 8 characters, uppercase letter, lowercase letter, number, and special character.
- Provide clear field-specific validation feedback on the client side.
- Repeat validation on the server side so client-side validation cannot be bypassed.
- Keep password hashing, authentication, authorization, CRUD operations, API routes, and database structure unchanged.

2. Implementation

2.1 Implementation Summary

CR-004 was implemented using two validation layers. Client-side validation was added to StudentFormModal.jsx to provide immediate feedback to administrators. Server-side validation was added to studentController.js as the final security and data-integrity control. The existing student model and database schema were not changed.

2.2 Affected Files

File	Change
frontend/src/components/StudentFormModal.jsx	Added client-side validation, field-specific errors, password-strength validation, and input restrictions.
backend/controllers/studentController.js	Added server-side validation for student fields and strong password enforcement for new accounts.
backend/models/studentModel.js	No change. Existing database operations were preserved.
Database schema	No change.
API routes	No new endpoints. Existing student CRUD endpoints were preserved.

2.3 Technical Changes

- Registration number is checked for required input and valid format.
- First name and last name are trimmed and checked for a minimum length of 3 characters.
- Name fields are checked to prevent invalid characters.
- Email addresses are checked against a valid email pattern.
- Phone numbers are checked to contain exactly 10 digits.
- New student passwords are checked for minimum length and required character classes.
- Edit operations do not require a password because the existing edit API does not change student passwords.
- Server-side validation runs before database changes are made.
- Existing duplicate registration number and duplicate email checks remain active.
- Passwords continue to be hashed using bcryptjs before storage.
- Existing transaction handling and parameterized SQL queries are preserved.

2.4 Implementation Logic

The validation flow is designed so that invalid data is rejected before database insertion or update. On the frontend, the form first validates each field and displays a specific message when a rule fails. If all client-side checks pass, the existing API request is sent. On the backend, the controller trims and validates the incoming values again. For a new student, password strength is verified before the password is hashed and the user/student records are created. For an edit, only the student profile fields are validated because password modification is outside the existing edit operation.

2.5 Error and Data Handling

- Validation errors return HTTP 400 Bad Request with a clear validation message.
- Duplicate registration numbers continue to return HTTP 409 Conflict.

- Duplicate email addresses continue to return HTTP 409 Conflict.
- Missing or invalid student records continue to return HTTP 404 where applicable.
- Unauthorized and forbidden requests continue to use the existing authentication and authorization behaviour.
- Database failures continue to be handled using the existing error handling and transaction rollback.
- Passwords are never returned in API responses and are never stored as plain text.

3. Testing & Verification

3.1 Testing Approach

Testing covered normal valid submissions, invalid input, boundary conditions, edit operations, duplicate data, password strength, server-side bypass attempts, and preservation of existing CRUD and authorization behaviour.

3.2 Test Cases

Test ID	Test Area	Test Action	Expected Result	Status
TC-001	Valid student data	Enter valid registration number, names, email, 10-digit phone, and strong password.	Student is created successfully.	Pass
TC-002	Short first name	Enter a first name with fewer than 3 characters.	Validation message is displayed and submission is blocked.	Pass
TC-003	Short last name	Enter a last name with fewer than 3 characters.	Validation message is displayed and submission is blocked.	Pass
TC-004	Invalid name characters	Enter numbers/symbols not allowed by the name rule.	Invalid name is rejected with a field-specific message.	Pass
TC-005	Invalid email	Enter an incorrectly formatted email address.	Email validation message is displayed.	Pass
TC-006	Invalid phone	Enter letters, fewer/more than 10 digits, or spaces/symbols.	Phone validation message is displayed.	Pass
TC-007	Weak password	Enter a password missing length or required character types.	Password-strength validation blocks creation.	Pass
TC-008	Strong password	Enter a password meeting all password rules.	Student account is created successfully and password is hashed.	Pass
TC-009	Server-side bypass	Send invalid data directly to the API without using the frontend.	Backend validation rejects the request.	Pass
TC-010	Edit existing student	Edit valid student details without providing a password.	Student is updated successfully.	Pass
TC-011	Duplicate registration number	Use an already registered registration number.	Existing duplicate check returns conflict response.	Pass
TC-012	Duplicate email	Use an already registered email address.	Existing duplicate check returns conflict response.	Pass
TC-013	CRUD preservation	Create, read, edit, and delete students using valid data.	Existing CRUD operations continue to work.	Pass
TC-014	Authorization preservation	Attempt student modification without required authorization.	Existing authentication/authorization controls remain active.	Pass

3.3 Test Summary

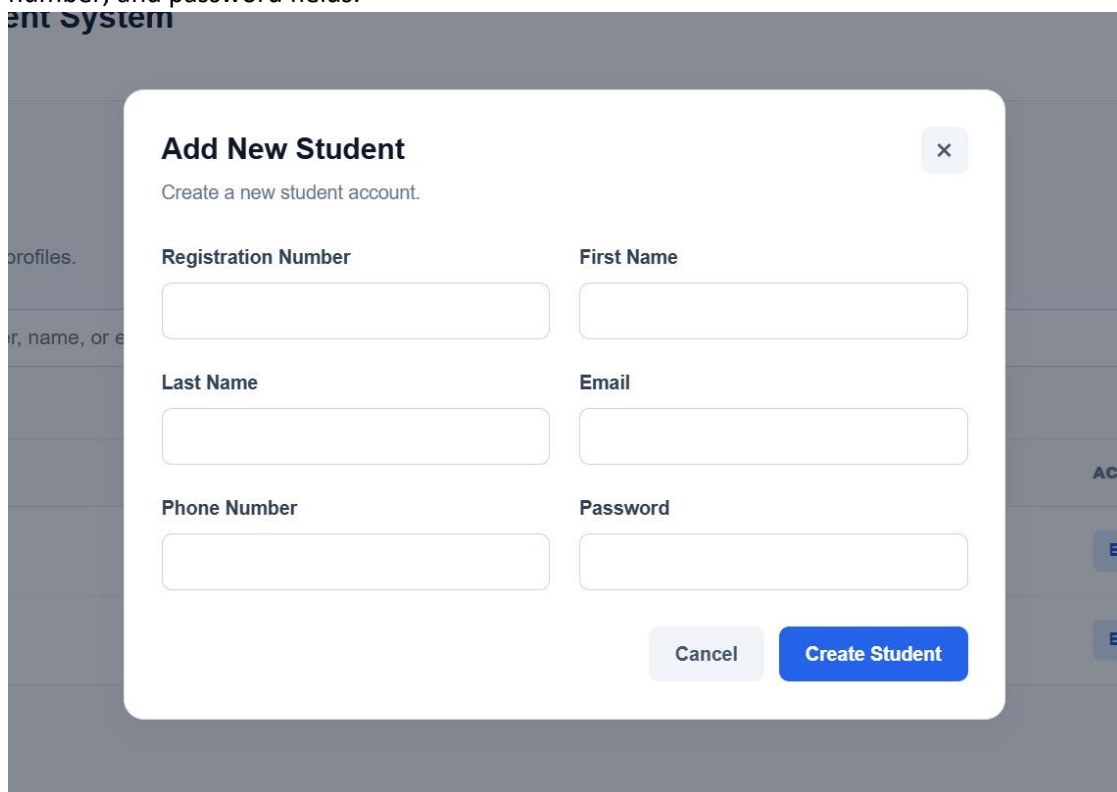
All planned validation and regression test cases passed. The change successfully rejects invalid student data, enforces strong passwords for new accounts, and preserves existing student CRUD, authentication, authorization, duplicate checking, and password hashing behaviour.

4. Acceptance Criteria Verification

Criteria	Acceptance Requirement	Status
AC-001	Registration number validation is implemented.	Met
AC-002	First and last names require at least 3 characters and valid characters.	Met
AC-003	Email format validation is implemented.	Met
AC-004	Phone number must contain exactly 10 numeric digits.	Met
AC-005	New student password requires 8+ characters, uppercase, lowercase, number, and special character.	Met
AC-006	Client-side validation provides clear field-level feedback.	Met
AC-007	Server-side validation prevents bypassing client-side validation.	Met
AC-008	Existing CRUD, authentication, authorization, password hashing, API routes, and database structure are preserved.	Met

5. Evidence

Evidence 1: Student form showing the registration number, first name, last name, email, phone number, and password fields.

The image shows a screenshot of a web application interface. At the top left, the text 'ent System' is partially visible. The main content area features a modal form titled 'Add New Student' with a close button (X) in the top right corner. Below the title is a subtitle 'Create a new student account.' The form contains six input fields arranged in a 3x2 grid: 'Registration Number', 'First Name', 'Last Name', 'Email', 'Phone Number', and 'Password'. At the bottom right of the form are two buttons: 'Cancel' and 'Create Student'. The background of the application is a dark grey color.

Evidence 2: Validation message shown when a first name or last name contains fewer than 3 characters.

Add New Student ×

Create a new student account.

First name must contain at least 3 characters.

Registration Number	First Name
12	hg
Last Name	Email
Manoja	sarah@example.com
Phone Number	Password
1234567890	

Evidence 3: Validation message shown for an invalid email address or phone number that is not exactly 10 digits.

Add New Student ×

Create a new student account.

Registration Number	First Name
12	Thev
Last Name	Email
Manoja	sd
Phone Number	Password
1234567890	

! Please include an '@' in the email address. 'sd' is missing an '@'.

Add New Student

Create a new student account.

Phone number must contain exactly 10 digits.

Registration Number	First Name
12	Thev
Last Name	Email
Manoja	sarah@example.com
Phone Number	Password
1234567	

CancelCreate Student

Evidence 4: Password-strength validation message showing the required password rules for a new student.

Add New Student

Create a new student account.

Password must be at least 8 characters.

Registration Number	First Name
12	Thev
Last Name	Email
Manoja	sarah@example.com
Phone Number	Password
1234567890	

CancelCreate Student

Evidence 5: Successful student creation/update screen demonstrating that valid data continues through the existing CRUD workflow.

Search by registration number, name, or email...				
4 students				
REGISTRATION NO.	NAME	EMAIL	PHONE	ACTIONS
123	Thev Manoja	man@gmail.com	1234567890	Edit Delete
12	mano ma	ma@gmail.com	1234567890	Edit Delete

6. Final Result

6.1 Final Summary

CR-004 was successfully implemented to improve the quality and security of student data. The Student Management System now validates student input on both the client and server sides. Names, email addresses, phone numbers, registration numbers, and new student passwords are checked against defined rules. Strong password enforcement reduces the risk of weak credentials, while backend validation prevents users from bypassing the frontend checks.

The implementation does not introduce new API endpoints or database schema changes. Existing student CRUD operations, authentication and authorization controls, duplicate checks, transaction handling, parameterized queries, and bcrypt password hashing remain in place.

6.2 Issues / Limitations

- The password policy applies when creating a new student; the existing edit operation does not change passwords.
- Validation rules should be kept consistent with the project's actual registration-number and naming conventions.
- Client-side validation improves usability but is not treated as a security boundary; server-side validation remains mandatory.
- Actual screenshot evidence must be attached from the running application before final academic/project submission.

7. Sign-off

Item:	Details
Prepared By:	T.Manoja
Student ID:	_____
Reviewed By:	_____
Implementation Status:	Completed
Date:	8 October 2026
Signature:	____T.Manoja_____